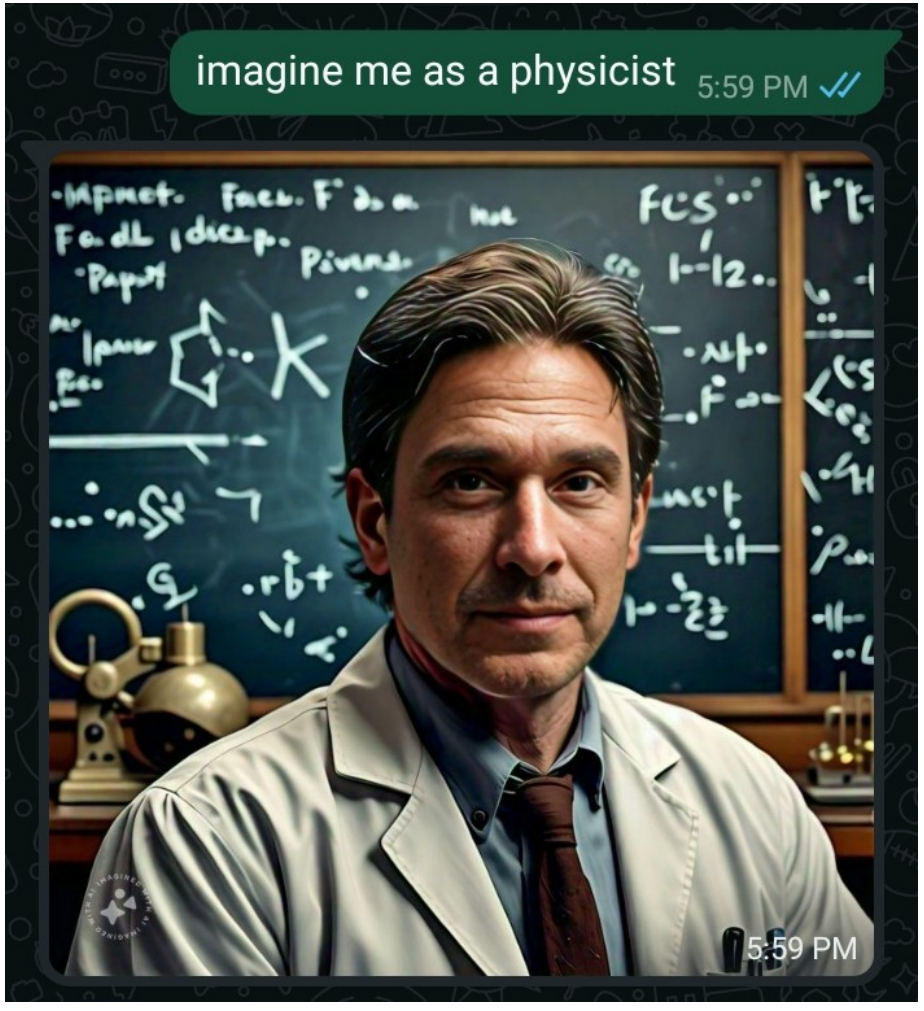



# Dealing with Real Data, Nearest Neighbor Algorithm

## Lecture/Lab 23
## PHYS 4730 F26

# Outline

- Review last time
  - Decision Trees (DT) in **scikit-learn**
  - Skepticism towards ML outcomes
- Real Data Sets
  - Notebook: Full HWC dataset
- The k Nearest Neighbor (kNN) Algorithm
  - Mini Lecture
  - Notebook: kNN in **scikit-learn**
- Lab 23: kNN on full HWC data



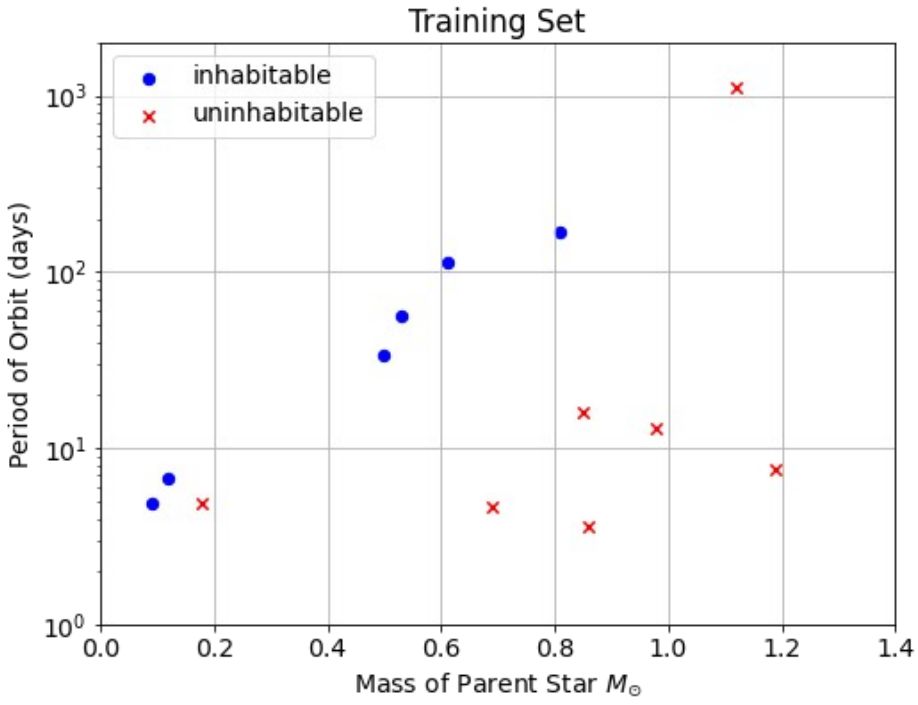

**Trees and Neighbors**

# Last Time: Subset of Habitable Worlds Catalog

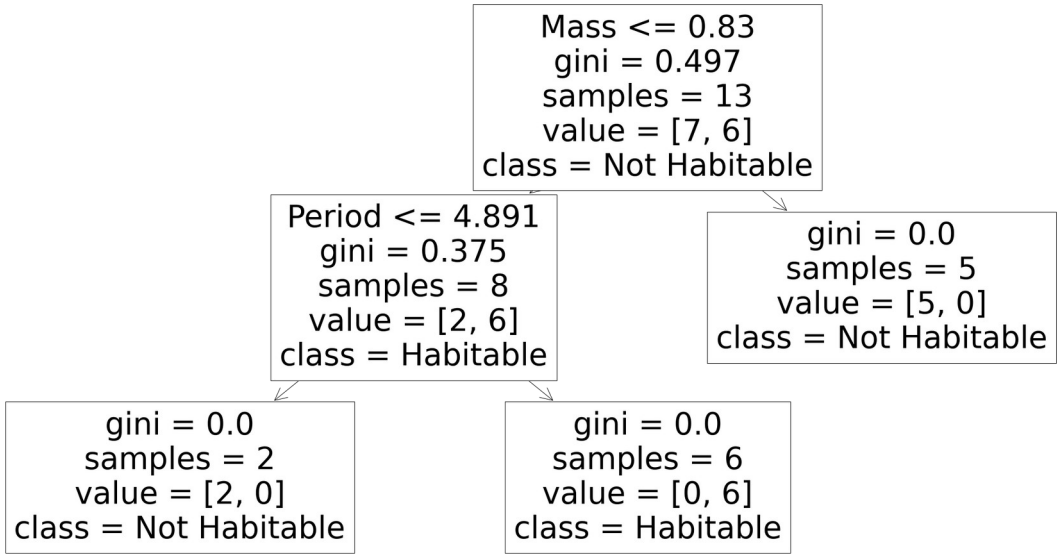
Training Set

Test Set

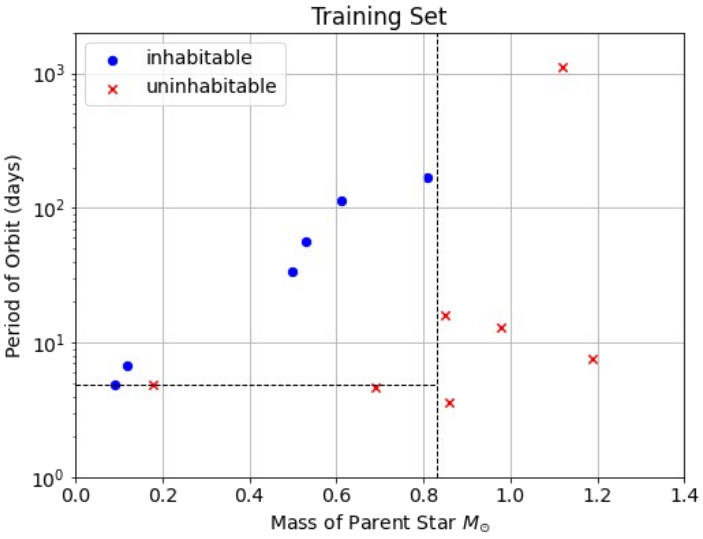
| | P_NAME | S_MASS | P_PERIOD | P_DISTANCE | P_HABITABLE |
|---|---|---|---|---|---|
| 0 | Kepler-736 b | 0.86 | 3.601472 | 0.043718 | 0 |
| 1 | Kepler-636 b | 0.85 | 16.080661 | 0.118082 | 0 |
| 2 | Kepler-887 c | 1.19 | 7.638460 | 0.080420 | 0 |
| 3 | Kepler-442 b | 0.61 | 112.305300 | 0.409327 | 1 |
| 4 | Kepler-772 b | 0.98 | 12.992073 | 0.107408 | 0 |
| 5 | Teegarden's Star b | 0.09 | 4.910000 | 0.025200 | 1 |
| 6 | K2-116 b | 0.69 | 4.655411 | 0.048086 | 0 |
| 7 | GJ 1061 c | 0.12 | 6.689000 | 0.035000 | 1 |
| 8 | HD 68402 b | 1.12 | 1103.000000 | 2.180981 | 0 |
| 9 | Kepler-1544 b | 0.81 | 168.811170 | 0.557107 | 1 |
| 10 | Kepler-296 e | 0.50 | 34.142110 | 0.178202 | 1 |
| 11 | Kepler-705 b | 0.53 | 56.056054 | 0.231926 | 1 |
| 12 | Kepler-445 c | 0.18 | 4.871229 | 0.031746 | 0 |
| 13 | HD 104067 b | 0.62 | 55.806000 | 0.260000 | 0 |
| 14 | GJ 4276 b | 0.41 | 13.352000 | 0.087613 | 0 |
| 15 | Kepler-296 f | 0.50 | 63.336270 | 0.268885 | 1 |
| 16 | Kepler-63 b | 0.98 | 9.434150 | 0.088100 | 0 |
| 17 | GJ 3293 d | 0.42 | 48.134500 | 0.195336 | 1 |

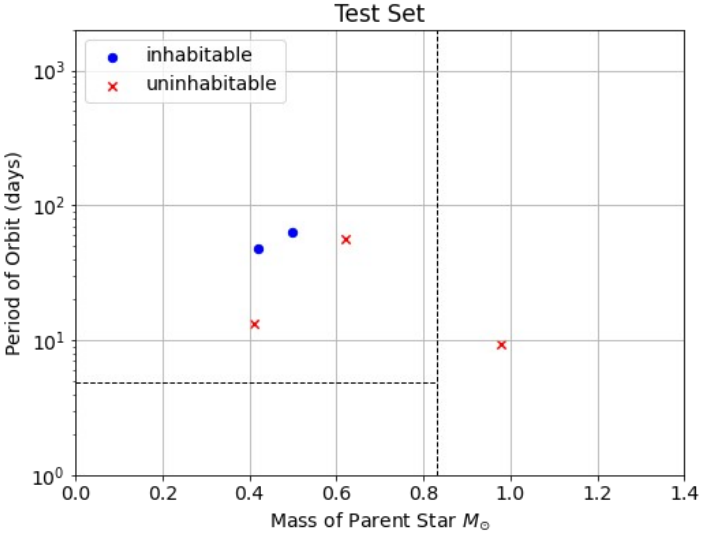


# Initial training set, cuts on mass, period

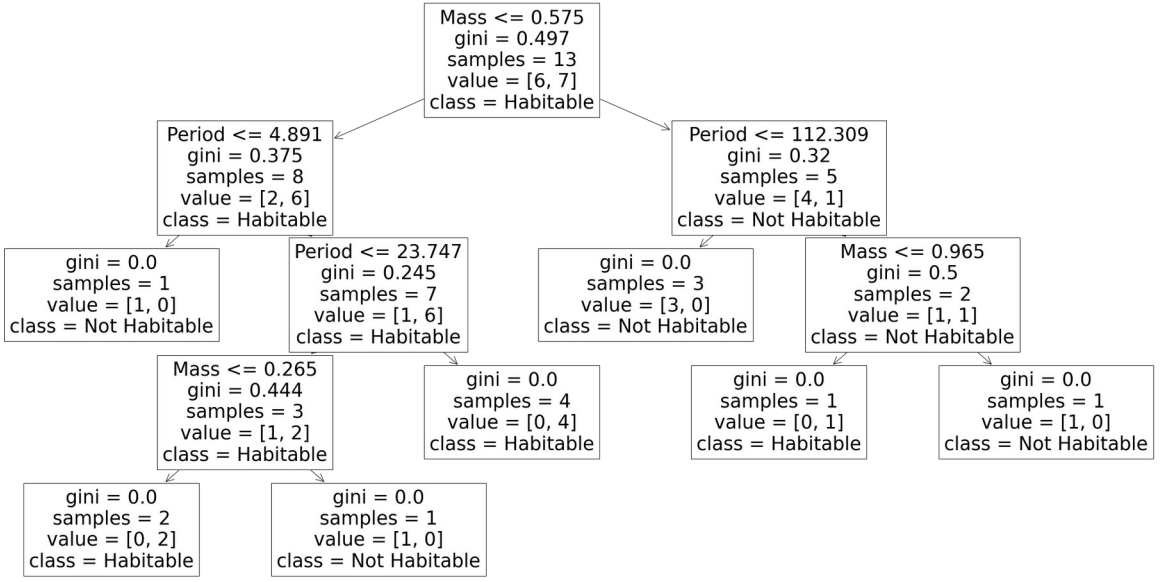



**Training accuracy score: 1.00**
**Test accuracy score: 0.60**

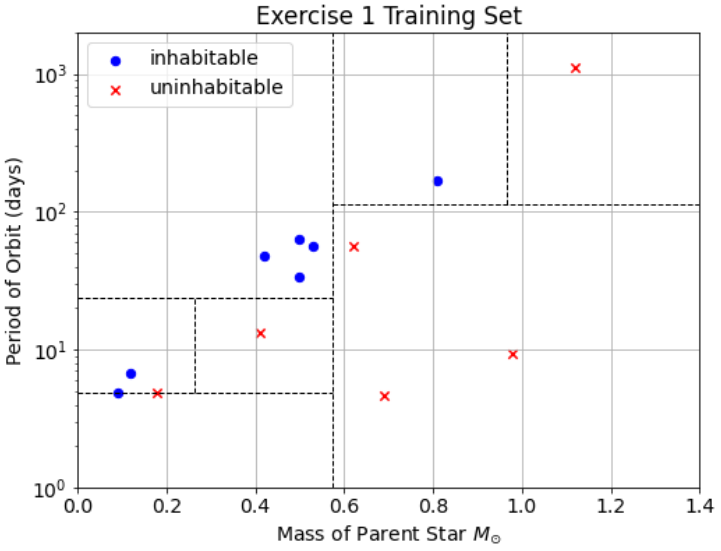


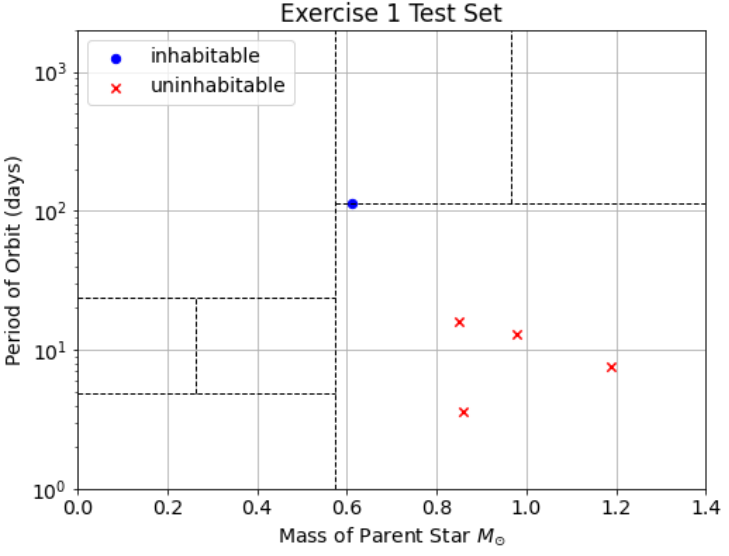



# 2nd training set, cuts on mass, period

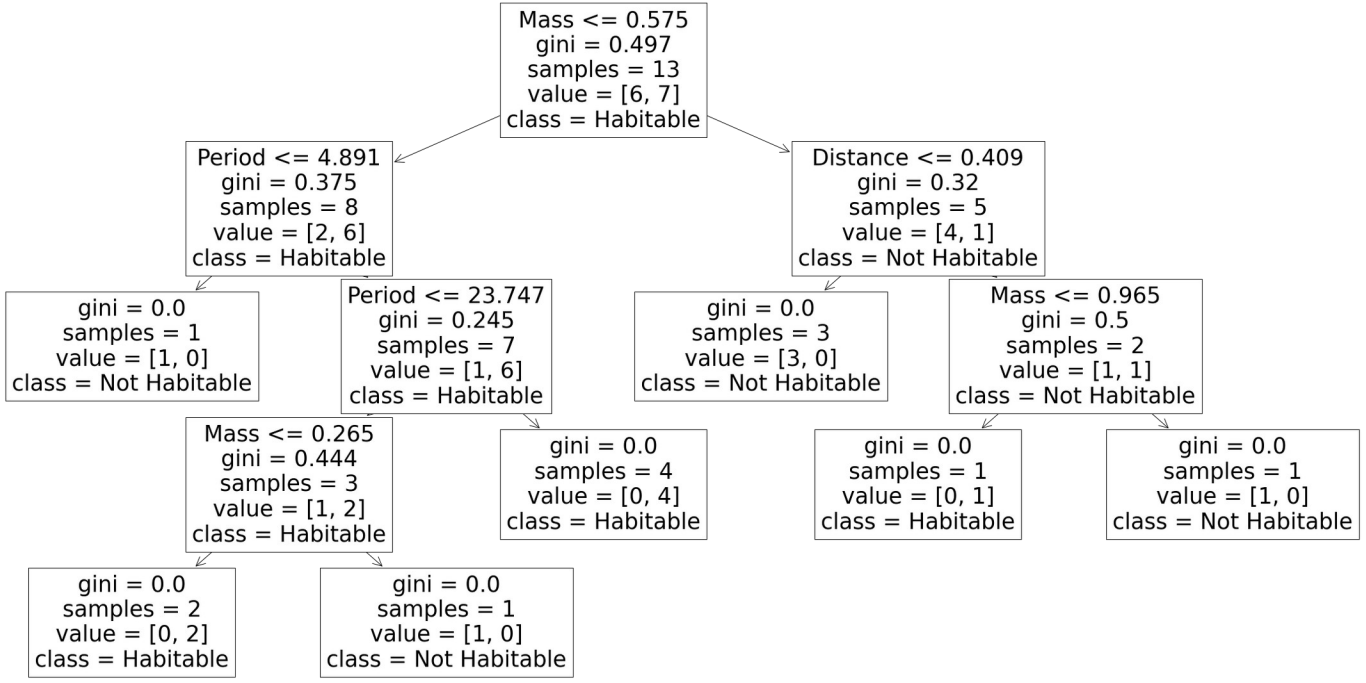



**Training accuracy score: 1.00**
**Test accuracy score: 0.80**



# 2nd training set, cuts on mass, period, distance




**Training accuracy score: 1.00**
**Test accuracy score: 1.00**

# Ex 3: Review Questions

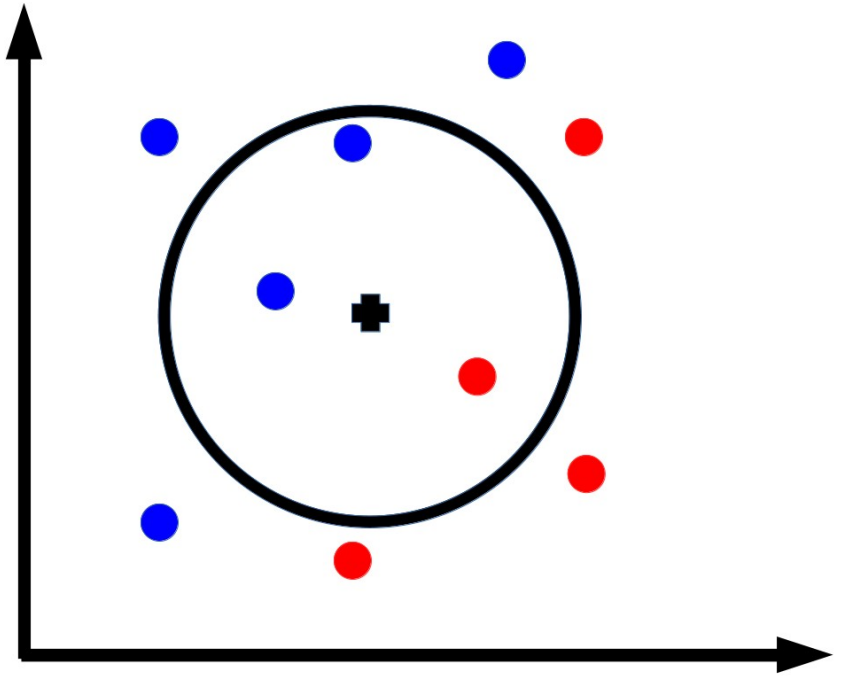
- The DT algorithm has as an advantage the fact that it is easy to understand and interpret. Can you think of any disadvantage(s)?
- Comment on the suitability of the data we have chosen for this lab. How might the dataset be improved?
- Would Earth be classified as a habitable planet?

# Dealing with Real Data

real_hwc.ipynb

# The *k*-Nearest Neighbor (kNN) Algorithm

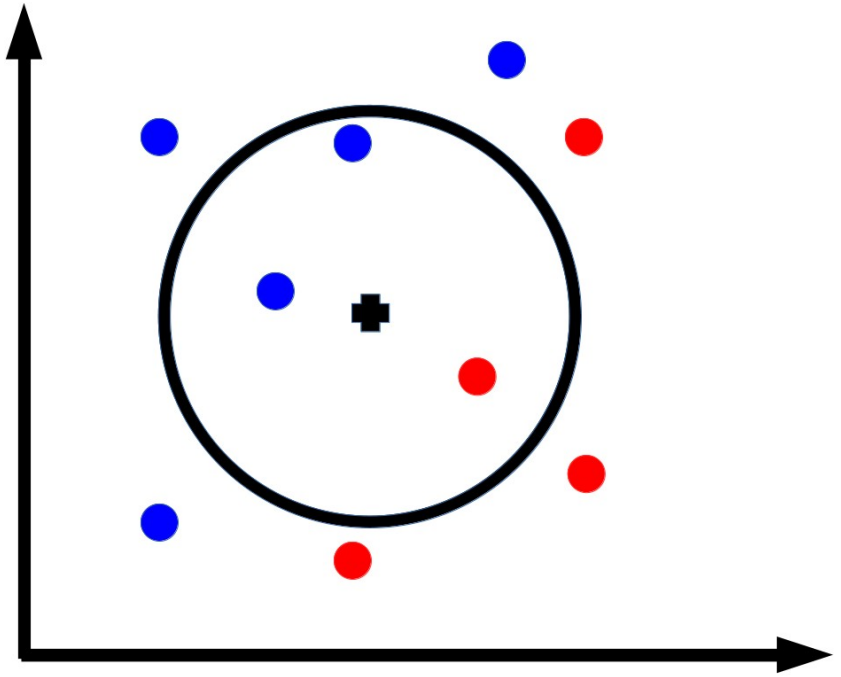
- When asked to classify an instance, the algorithm will look for the *k* (with *k* being a predetermined integer) closest examples in feature space in the training set, and report as a decision the predominant class among those examples.



***k* = 3**
**Cross is assigned label “blue”**

# The *k*-Nearest Neighbor (kNN) Algorithm

- Variants:
  - Different integers *k*
  - Weighing training events, e.g. by $1/r^2$.
- Caveats:
  - Handling “ties” (e.g. k-even with binary classifier).
  - Features may have different ranges, so need to rescale for distance calculation.



***k* = 3**
**Cross is assigned label “blue”**

# *k*-Nearest Neighbor (kNN) Algorithm in scikit-learn

knn_sk.ipynb

PHYSICS 4730
Lab 23

- Exercises
- Assignment

---

**Exercise 1.**

Create a Python script or Jupyter Notebook to read and apply the kNN classification algorithm to the full Habitable Worlds Catalog data set. For starters, just use two features (*e.g.* stellar mass and orbital period) for the classification. Be sure that the training data is appropriately scaled for use with kNN.

If you are able to get that working, here are some variants to try:

- Try adding additional features to the classifier. Are there any that we have not tried in lab or lecture that you think may be relevant?
- Try varying *k*. Does it make any difference in the performance?
- Instead of weighing each of the *k* nearest neighbors uniformly, it is possible to weigh them according to distance or a user-defined function. Can you get this functionality to work?
- To think about: How are kNN and DT similar and different in performance? Are there cases where one or the other would be clearly preferable? Next week we will begin looking at tools to provide quantitative answers to these questions.

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**There are no required submissions for today. Any code created will be applicable to Homework #5.**

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Back to main Physics 4730 page